CSE 4630: Containerization for Developers

Department of Computer Science and Information Technology

7th Semester

Assignment 2

Docker Images and Dockerfiles

Book Reference	Chapters 4–5

Practical Questions

Q. No.	Question
1	Create a Dockerfile for a small Python application that reads student marks from a file and displays the average, grade, and pass percentage. Build and run the image.
	Modify the application so that the application name and passing percentage are supplied through environment variables rather than being hard-coded.
	Build an image named my-python-app:v1 from the Dockerfile and run a container from it.
2	Create a Compose application containing a frontend web server and a backend service. Configure them so that the frontend communicates with the backend using the Compose service name rather than an IP address.
	Modify the above application so that generated reports are stored in a named volume and remain available after the containers are recreated.
3	Create a Compose configuration in which the database password, application port, and database name are supplied through environment variables.
	Intentionally stop the backend service while the frontend is running. Investigate the resulting behavior using Docker commands and restore the application without recreating unrelated services.
4	Design and run a three-service application consisting of a web server, application service, and database. Provide the Dockerfile(s), compose.yml, volume configuration, network configuration, and commands required to start, stop, inspect, and remove the complete application.